

Section A

Question 1

Solution

Let $(\Omega, \mathcal{F}, P)$ be the given probability space.

W.K.T,

$$\sum_{n=1}^{\infty} P(A_n) < \infty.$$

We need to show that the probability of infinitely many of the events A_n happening is equal to 0.

Lets take,

$$E = \limsup_{n \rightarrow \infty} A_n.$$

Now

here we defined E as,

$$E = \bigcap_{m=1}^{\infty} \bigcup_{n=m}^{\infty} A_n.$$

Let

$$E_m = \bigcup_{n=m}^{\infty} A_n.$$

$$E = \bigcap_{m=1}^{\infty} E_m.$$

By observing , E_m means that at least one of the events $A_m, A_{m+1}, A_{m+2}, \dots$ happening.

Also by observation,

$$E_{m+1} \subseteq E_m$$

as m increases there are less events inside the union.

Therefore E_m is a decreasing seq.

Now, W.K.T

$$P(E) = P\left(\bigcap_{m=1}^{\infty} E_m\right) = \lim_{m \rightarrow \infty} P(E_m).$$

By Boole's inequality,

$$P(E_m) = P\left(\bigcup_{n=m}^{\infty} A_n\right) \leq \sum_{n=m}^{\infty} P(A_n).$$

Hence,

$$P(E) = \lim_{m \rightarrow \infty} P(E_m) \leq \lim_{m \rightarrow \infty} \sum_{n=m}^{\infty} P(A_n).$$

Since W.K.T the series

$$\sum_{n=1}^{\infty} P(A_n)$$

is converging, the R.H.S sums tends to 0 as $m \rightarrow \infty$.

Hence,

$$\lim_{m \rightarrow \infty} \sum_{n=m}^{\infty} P(A_n) = 0.$$

This gives

$$P(E) \leq 0.$$

But a probability can never be $-ve$, therefore

$$P(E) = 0.$$

Hence,

$$P(\limsup A_n) = 0.$$

Hence , the probability that infinitely many of the events occur is 0.

From the Above arguments;

Therefore,

$$\boxed{P(A_n \text{ infinitely often}) = 0}$$

Hence proved

Question 2

Solution

Using De Morgan's law,

$$\left(\bigcap_{i=1}^n A_i\right)^c = \bigcup_{i=1}^n A_i^c.$$

Hence,

$$P\left(\bigcap_{i=1}^n A_i\right) = 1 - P\left(\bigcup_{i=1}^n A_i^c\right).$$

By Boole's inequality,

$$P\left(\bigcup_{i=1}^n A_i^c\right) \leq \sum_{i=1}^n P(A_i^c).$$

Therefore,

$$P\left(\bigcap_{i=1}^n A_i\right) \geq 1 - \sum_{i=1}^n P(A_i^c).$$

Since W.K.T,

$$P(A_i^c) = 1 - p_i,$$

we get

$$P\left(\bigcap_{i=1}^n A_i\right) \geq 1 - \sum_{i=1}^n (1 - p_i).$$

Also probability can never be $-ve$, so

$$P\left(\bigcap_{i=1}^n A_i\right) \geq 0.$$

Combining both results,

$$P\left(\bigcap_{i=1}^n A_i\right) \geq \max\left\{1 - \sum_{i=1}^n (1 - p_i), 0\right\}.$$

Hence from above statements we get ,

$$\boxed{P\left(\bigcap_{i=1}^n A_i\right) \geq \max\left\{\sum_{i=1}^n p_i - (n - 1), 0\right\}}$$

Question 3

Solution

Given,

$$P(A|B) \geq P(A).$$

Therefore,

$$\frac{P(A \cap B)}{P(B)} \geq P(A),$$

which implies

$$P(A \cap B) \geq P(A)P(B). \quad (1)$$

Assumption:

$$P(A^c|B^c) < P(A^c).$$

Then

$$\frac{P(A^c \cap B^c)}{P(B^c)} < P(A^c),$$

and hence

$$P(A^c \cap B^c) < P(A^c)P(B^c). \quad (2)$$

Now W.K.T,

$$P(A^c \cap B^c) = 1 - P(A) - P(B) + P(A \cap B),$$

Substituting values in 2;

$$1 - P(A) - P(B) + P(A \cap B) < (1 - P(A))(1 - P(B)).$$

Finally , we get

$$P(A \cap B) < P(A)P(B),$$

The above statement derived contradicts our assumption

Therefore our assumption is false.

Hence,

$$P(A^c|B^c) \geq P(A^c).$$

Hence from above statements using original identity(without complimentary functions)

$$\boxed{P(A|B) \geq P(A) \implies P(A^c|B^c) \geq P(A^c)}$$

Section B

Question 4

Solution

Let

$$P(N) = \frac{\binom{T}{k} \binom{N-T}{n-k}}{\binom{N}{n}}.$$

Consider

$$\frac{P(N)}{P(N-1)} = \frac{N-T}{N-T-(n-k)} \cdot \frac{N-n}{N}.$$

Thus

$$\frac{P(N)}{P(N-1)} = \frac{(N-T)(N-n)}{N(N-n-T+k)}.$$

Setting

$$\frac{P(N)}{P(N-1)} \geq 1,$$

gives

$$(N-T)(N-n) \geq N(N-n-T+k).$$

$$nT \geq kN.$$

$$N \leq \frac{nT}{k}.$$

Hence $P(N)$ increases for

$$N \leq \frac{nT}{k}$$

and decreases for

$$N > \frac{nT}{k}.$$

Hence

$$\hat{N} = \left\lfloor \frac{nT}{k} \right\rfloor.$$

Question 5

Solution

$$P(X = k) = \frac{\binom{n}{k} D_{n-k}}{n!}.$$

W.K.T,

$$D_m = m! \sum_{j=0}^m \frac{(-1)^j}{j!}.$$

$$\Rightarrow D_{n-k} = (n-k)! \sum_{j=0}^{n-k} \frac{(-1)^j}{j!}.$$

$$P(X = k) = \frac{\binom{n}{k} (n-k)!}{n!} \sum_{j=0}^{n-k} \frac{(-1)^j}{j!}.$$

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

$$\Rightarrow P(X = k) = \frac{1}{k!} \sum_{j=0}^{n-k} \frac{(-1)^j}{j!}.$$

Now,

$$e^{-1} = \sum_{j=0}^{\infty} \frac{(-1)^j}{j!}.$$

$$\left| e^{-1} - \sum_{j=0}^{n-k} \frac{(-1)^j}{j!} \right| \leq \sum_{j=n-k+1}^{\infty} \frac{1}{j!}.$$

$$\sum_{j=n-k+1}^{\infty} \frac{1}{j!} \leq \frac{e}{(n-k+1)!} \rightarrow 0.$$

$$\Rightarrow \sum_{j=0}^{n-k} \frac{(-1)^j}{j!} \rightarrow e^{-1}.$$

$$\Rightarrow P(X = k) = \frac{1}{k!} \sum_{j=0}^{n-k} \frac{(-1)^j}{j!} \rightarrow \frac{1}{k!} e^{-1}.$$

$$\boxed{\lim_{n \rightarrow \infty} P(X = k) = \frac{1}{e k!}}$$

Conclusion

$$P(X = k) = \frac{\binom{n}{k} D_{n-k}}{n!}$$

$$\lim_{n \rightarrow \infty} P(X = k) = \frac{1}{e k!}$$

Question 6

Solution

$$P(R_2) = P(R_2|R_1)P(R_1) + P(R_2|B_1)P(B_1).$$

$$P(R_1) = \frac{r}{r+b}, \quad P(B_1) = \frac{b}{r+b}.$$

Using CP, If first draw is red,

$$P(R_2|R_1) = \frac{r}{r+b+c}.$$

If first draw is blue,

$$P(R_2|B_1) = \frac{r+c}{r+b+c}.$$

Hence

$$\begin{aligned} P(R_2) &= \frac{r}{r+b+c} \frac{r}{r+b} + \frac{r+c}{r+b+c} \frac{b}{r+b} \\ &= \frac{r(r+b+c)}{(r+b)(r+b+c)} = \frac{r}{r+b}. \end{aligned}$$

Therefore

$$P(R_2) = \frac{r}{r+b}.$$

For R_3 ,

$$P(R_3) = P(R_3|R_2)P(R_2) + P(R_3|B_2)P(B_2).$$

Since

$$P(R_2) = \frac{r}{r+b}, \quad P(B_2) = \frac{b}{r+b},$$

and

$$P(R_3|R_2) = \frac{r}{r+b+c},$$

$$P(R_3|B_2) = \frac{r+c}{r+b+c},$$

we get

$$\begin{aligned} P(R_3) &= \frac{r}{r+b+c} \frac{r}{r+b} + \frac{r+c}{r+b+c} \frac{b}{r+b} \\ &= \frac{r}{r+b}. \end{aligned}$$

Thus

$$\boxed{P(R_3) = \frac{r}{r+b}.$$

Similarly, if

$$P(R_n) = \frac{r}{r+b},$$

then

$$P(R_{n+1}) = P(R_{n+1}|R_n)P(R_n) + P(R_{n+1}|B_n)P(B_n)$$

gives

$$P(R_{n+1}) = \frac{r}{r+b}.$$

Hence

$$\boxed{P(R_n) = \frac{r}{r+b} \quad \forall n \geq 1.}$$

Therefore the probability of drawing a red ball is time-invariant even under this modified replacement rule.

Section C

Question 7

Solution

Let

$$W = \{\text{warden says "B will be executed"}\}.$$

$$P(A) = P(B) = P(C) = \frac{1}{3}.$$

$$P(W|A) = \frac{1}{2}, \quad P(W|B) = 0, \quad P(W|C) = 1.$$

$$P(W) = P(W|A)P(A) + P(W|B)P(B) + P(W|C)P(C)$$

$$= \frac{1}{2} \cdot \frac{1}{3} + 0 \cdot \frac{1}{3} + 1 \cdot \frac{1}{3}$$

$$= \frac{1}{6} + \frac{1}{3} = \frac{1}{2}.$$

$$\begin{aligned} P(A|W) &= \frac{P(W|A)P(A)}{P(W)} \\ &= \frac{\frac{1}{2} \cdot \frac{1}{3}}{\frac{1}{2}} \\ &= \frac{1}{3}. \end{aligned}$$

Conclusion

$$P(A|W) = \frac{1}{3}$$

$\therefore$

Prisoner A is incorrect.

$$P(\text{A survives}) = \frac{1}{3} \neq \frac{1}{2}.$$

Question 8

Solution

Let A be the event that Alice wins.

Alice can win on her first toss, or after both players have tossed tails once, or after both players have tossed tails twice, etc

Hence

$$P(A) = p + (1-p)^2p + (1-p)^4p + \dots$$

$$= p \sum_{n=0}^{\infty} (1-p)^{2n}.$$

Using the geometric series formula,

$$\begin{aligned} P(A) &= \frac{p}{1 - (1-p)^2} \\ &= \frac{p}{2p - p^2} \\ &= \frac{1}{2 - p}. \end{aligned}$$

Therefore

$$P(A) = \frac{1}{2-p}$$

and

$$P(A^c) = 1 - P(A) = \frac{1-p}{2-p}.$$

For the game to be fair,

$$P(A) = P(A^c).$$

Thus

$$\frac{1}{2-p} = \frac{1-p}{2-p}.$$

$$1 = 1 - p.$$

$$p = 0.$$

But

$$p \in (0, 1).$$

Therefore no value of p in $(0, 1)$ makes the game fair.

Also,

$$2 - p < 2$$

for all $p \in (0, 1)$, so

$$P(A) = \frac{1}{2-p} > \frac{1}{2}.$$

Hence Alice always has an advantage because she tosses first.

$$P(A) > \frac{1}{2} \quad \forall p \in (0, 1).$$

There does not exist $p \in (0, 1)$ such that the game is perfectly fair.

Question 9

Problem Statement

If A, B, C are pairwise independent, must they be mutually independent? Prove or give a counterexample.

Solution

Lets take

$$(\Omega, \mathcal{F}, P)$$

, where every outcome is equally-likely occuring;

$$\Omega = \{a, b, c, d\}, \quad P(\Omega_i) = \frac{1}{4}.$$

$$A = \{a, b\}, \quad B = \{a, c\}, \quad C = \{a, d\}.$$

$$P(A) = P(B) = P(C) = \frac{1}{2}.$$

$$A \cap B = \{a\}$$

$$P(A \cap B) = \frac{1}{4} = P(A)P(B).$$

$$A \cap C = \{a\}$$

$$P(A \cap C) = \frac{1}{4} = P(A)P(C).$$

$$B \cap C = \{a\}$$

$$P(B \cap C) = \frac{1}{4} = P(B)P(C).$$

Hence,

A, B, C are pairwise independent.

$$A \cap B \cap C = \{a\}.$$

$$P(A \cap B \cap C) = \frac{1}{4}.$$

$$P(A)P(B)P(C) = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}.$$

$$\frac{1}{4} \neq \frac{1}{8}.$$

$$\boxed{P(A \cap B \cap C) \neq P(A)P(B)P(C)}$$

Hence , by above statements the condition doesn't necessarily imply that they are mutually independent.

Section D

Question 11

Problem Statement

Construct a valid CDF F such that X is discrete, its support is

$$\mathbb{Q} \cap [0, 1],$$

and F is strictly increasing on $(0, 1)$.

Solution

Enumerate

$$\mathbb{Q} \cap [0, 1] = \{q_1, q_2, \dots\}.$$

Define

$$P(X = q_n) = 2^{-n}.$$

Since

$$\sum_{n=1}^{\infty} 2^{-n} = 1,$$

this is a valid probability distribution.

Set

$$F(x) = \sum_{n=1}^{\infty} \frac{1}{2^n} \mathbf{1}_{\{q_n \leq x\}}.$$

Also,

$$F(-\infty) = 0, \quad F(+\infty) = 1.$$

Let

$$a < b.$$

Since $\mathbb{Q}$ is dense in $\mathbb{R}$,

$$\exists q_n \in (a, b).$$

Hence

$$F(b) - F(a) \geq 2^{-n} > 0.$$

Therefore,

$$F(b) > F(a).$$

$F \text{ is strictly increasing on } (0, 1).$

Conclusion

$$F(x) = \sum_{n=1}^{\infty} \frac{1}{2^n} \mathbf{1}_{\{q_n \leq x\}}$$

Question 12

Problem Statement

Let X be the distance from the origin to a point chosen uniformly from the unit disk

$$x^2 + y^2 \leq 1.$$

Find the CDF and PDF of X .

Solution

$$F_X(x) = P(X \leq x).$$

$$= \frac{\pi x^2}{\pi} = x^2, \quad 0 \leq x \leq 1.$$

$$F_X(x) = \begin{cases} 0, & x < 0, \\ x^2, & 0 \leq x \leq 1, \\ 1, & x > 1. \end{cases}$$

$$f_X(x) = \frac{d}{dx} F_X(x) = 2x.$$

$$f_X(x) = \begin{cases} 2x, & 0 < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Conclusion

$$F_X(x) = \begin{cases} 0, & x < 0, \\ x^2, & 0 \leq x \leq 1, \\ 1, & x > 1. \end{cases}$$

$$f_X(x) = \begin{cases} 2x, & 0 < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Question 13

Problem Statement

Let

$$X \sim U(-1, 1), \quad Y = X^2.$$

Find the CDF and PDF of Y .

Solution

$$\begin{aligned} F_Y(y) &= P(Y \leq y) = P(X^2 \leq y). \\ &= P(-\sqrt{y} \leq X \leq \sqrt{y}). \\ &= \frac{\sqrt{y} - (-\sqrt{y})}{2} = \sqrt{y}, \quad 0 \leq y \leq 1. \end{aligned}$$

$$F_Y(y) = \begin{cases} 0, & y < 0, \\ \sqrt{y}, & 0 \leq y \leq 1, \\ 1, & y > 1. \end{cases}$$

$$f_Y(y) = \frac{d}{dy} F_Y(y) = \frac{1}{2\sqrt{y}}.$$

$$f_Y(y) = \begin{cases} \frac{1}{2\sqrt{y}}, & 0 < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Conclusion

$$F_Y(y) = \begin{cases} 0, & y < 0, \\ \sqrt{y}, & 0 \leq y \leq 1, \\ 1, & y > 1. \end{cases}$$

$$f_Y(y) = \begin{cases} \frac{1}{2\sqrt{y}}, & 0 < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Section E

Question 15

Solution

Since X_1 and X_2 are independent,

$$P(X_1 = X_2) = \int_{-\infty}^{\infty} P(X_2 = x) dF(x).$$

Since F is continuous,

$$P(X_2 = x) = 0 \quad \forall x \in \mathbb{R}.$$

Hence

$$P(X_1 = X_2) = \int_{-\infty}^{\infty} 0 dF(x) = 0.$$

Therefore

$$\boxed{P(X_1 = X_2) = 0.}$$

Question 16

Solution

The CDF of X is

$$F(x) = \begin{cases} 0, & x < 0, \\ p, & x = 0, \\ p + (1 - p)G(x), & x > 0. \end{cases}$$

The q -th quantile ξ_q satisfies

$$F(\xi_q) \geq q.$$

If

$$0 < q \leq p,$$

then

$$\boxed{\xi_q = 0.}$$

If

$$p < q < 1,$$

then

$$p + (1 - p)G(\xi_q) = q,$$

which gives

$$G(\xi_q) = \frac{q - p}{1 - p}.$$

Hence

$$\xi_q = G^{-1}\left(\frac{q-p}{1-p}\right).$$

Therefore,

$$\xi_q = \begin{cases} 0, & 0 < q \leq p, \\ G^{-1}\left(\frac{q-p}{1-p}\right), & p < q < 1. \end{cases}$$